VIP C2 PROGRAMME — FULL STACK DEVELOPMENT WITH MERN

# Proposed Solution

Online Complaint Registration System — MERN Stack Complaint Management Platform

**Phase:** Brainstorming & Ideation Phase

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# Proposed Solution

## 1. System Context Diagram [cite: 1220]

The system context boundary establishes structural definitions mapping human actors, software platform hubs, and downstream cloud database components using a formalized C4 model protocol specification.

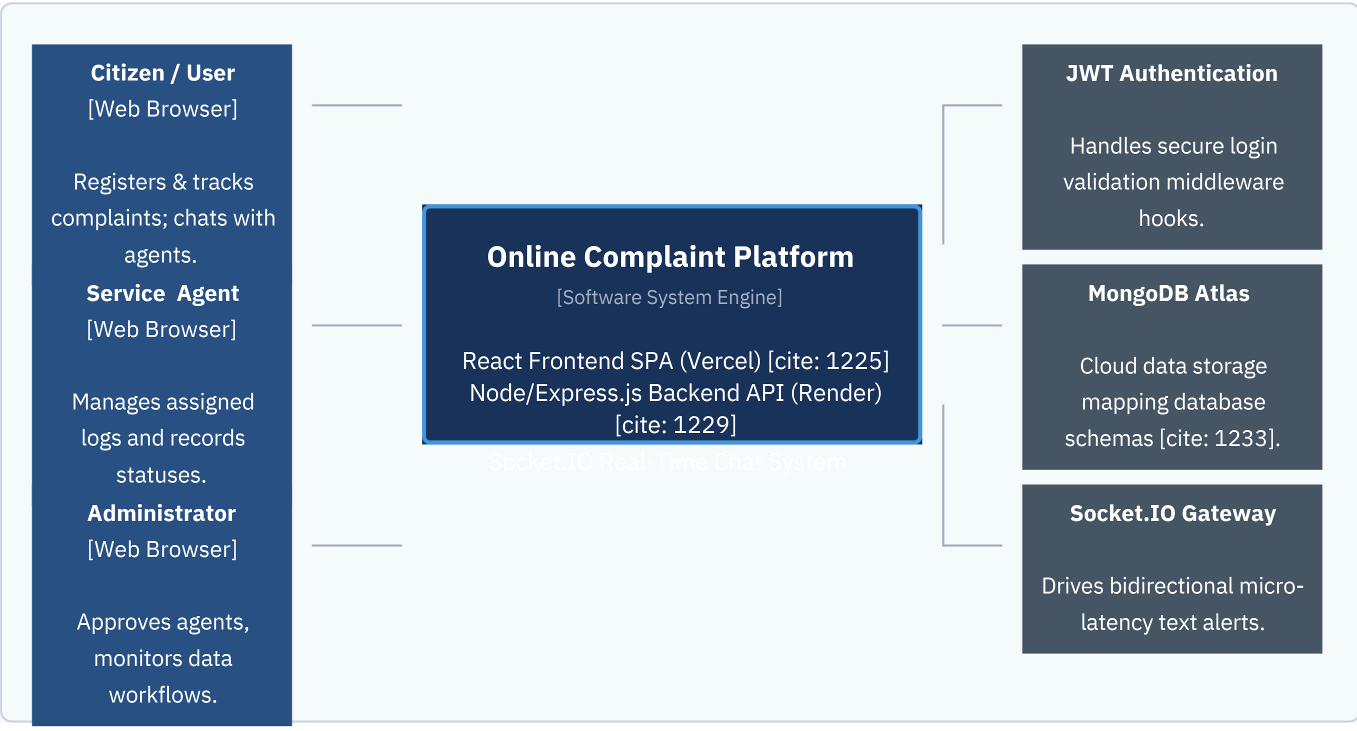


FIGURE 8.1 — C4 MODEL LEVEL 1: SYSTEM CONTEXT DIAGRAM [CITE: 1237]

## 2. Solution Parameters [cite: 1238]

TABLE 8.1 — SOLUTION PARAMETERS

| S.No | Parameter | Detailed Description |
|---|---|---|
| 1 | **Problem Statement** | Citizens often face difficulties in registering complaints, tracking progress, receiving updates, and communicating with responsible authorities. Traditional complaint management systems suffer from poor transparency, delayed responses, fragmented communication channels, and inefficient complaint handling processes. Administrators face challenges in monitoring complaint resolution activities, assigning complaints effectively, and generating meaningful performance insights. |

| S.No | Parameter | Detailed Description |
| --- | --- | --- |
| 2 | **Solution Description** | The Online Complaint Registration System is a full-stack MERN web application designed to streamline complaint management through a centralized digital platform. Citizens can register complaints, track complaint status, communicate with assigned agents through Socket.IO-powered real-time chat, receive notifications, and submit feedback after resolution. Agents can manage assigned complaints and update complaint statuses. Administrators can manage users, approve agents, assign complaints, monitor complaint progress, and analyze system-wide performance through analytics dashboards. |
| 3 | **Novelty / Uniqueness** | The solution combines complaint registration, complaint lifecycle management, real-time communication, notifications, analytics, and feedback collection within a single platform. The system leverages Socket.IO for instant communication, JWT-based authentication for secure access control, and role-based authorization supporting Citizens, Agents, and Administrators. Its cloud-based deployment using Render and Vercel enables accessibility and scalability. |
| 4 | **Social Impact / Customer Satisfaction** | The platform improves transparency, accountability, and responsiveness in complaint handling. Citizens gain confidence through visibility into complaint progress and receive timely updates. Agents benefit from streamlined workflows and centralized information. Administrators gain actionable insights through analytics dashboards. Overall, the system enhances trust, reduces complaint resolution time, and improves citizen satisfaction. |
| 5 | **Business Model (Revenue Model)** | The primary objective of the project is service improvement rather than revenue generation. However, future deployments may adopt Software-as-a-Service (SaaS) licensing for municipalities, educational institutions, corporate grievance systems, or public service organizations. Premium analytics and reporting features may also serve as potential revenue streams. |
| 6 | **Scalability of the Solution** | The MERN architecture supports horizontal scalability through separation of frontend and backend services. MongoDB Atlas enables scalable cloud database management, while Render and Vercel support independent deployment and scaling of application layers. The modular architecture facilitates future integration of AI-powered complaint classification, mobile applications, SMS notifications, and multilingual support. |

## 3. High-Level Feature Summary [cite: 1241]

**TABLE 8.2 — HIGH-LEVEL FEATURE SUMMARY**

| Feature Category | Key Capabilities |
| --- | --- |
| **Authentication & Security** | JWT Authentication, Password Encryption, Role-Based Access Control (Citizen / Agent / Admin) |
| **Complaint Registration** | Complaint Submission, Category Selection, Complaint Description, Complaint Tracking |

| Feature Category | Key Capabilities |
|---|---|
| **Complaint Lifecycle Management** | Complaint Assignment, Status Updates, Resolution Workflow, Complaint Monitoring |
| **Real-Time Communication** | Socket.IO Real-Time Chat between Citizens and Agents |
| **Notification System** | Real-Time Complaint Status Notifications, Assignment Notifications, Resolution Alerts |
| **Agent Operations** | Assigned Complaint Management, Status Updates, Citizen Communication |
| **Administrative Operations** | User Management, Agent Approval Workflow, Complaint Assignment, Complaint Monitoring |
| **Analytics Dashboard** | Complaint Statistics, Complaint Status Metrics, User Metrics, Agent Performance Monitoring |
| **Feedback Management** | Complaint Rating System, Citizen Feedback Collection, Satisfaction Monitoring |
| **Database Management** | MongoDB Atlas Storage for Users, Complaints, Messages, Notifications, Feedback |
| **Cloud Deployment** | Frontend Deployment on Vercel, Backend Deployment on Render |
| **Scalability & Extensibility** | Modular Architecture, Cloud Infrastructure, Future Feature Integration Support |

## 4. Proposed Benefits of the Solution

The proposed Online Complaint Registration System provides significant advantages over traditional complaint handling methods.

For citizens, the platform delivers a transparent and user-friendly environment where complaints can be submitted, monitored, and resolved efficiently. Real-time updates reduce uncertainty while direct communication with agents improves responsiveness.

For agents, the platform simplifies complaint management through centralized dashboards, assignment workflows, and communication tools. This reduces administrative overhead and improves operational efficiency.

For administrators, the platform offers centralized governance and analytics capabilities that support informed decision-making, performance monitoring, and process optimization.

The use of cloud infrastructure, secure authentication mechanisms, and modular architecture ensures reliability, maintainability, and future scalability.

## 5. Conclusion

The Online Complaint Registration System provides a comprehensive digital solution for modern complaint management requirements. By integrating complaint registration, assignment, tracking, communication, notifications, analytics, and feedback mechanisms into a unified platform, the system addresses the major challenges identified during the problem analysis phase.

The proposed solution demonstrates strong alignment with stakeholder requirements while leveraging modern web technologies such as MongoDB Atlas, Express.js, React.js, Node.js, Socket.IO, and JWT Authentication. The architecture supports future growth and provides a scalable foundation for enhancing public service delivery and citizen engagement.